

A Trait-level gate decisions

Amyloid- β 42

GCST90726396

1

APOE

regional + gene-based signal



1 APOE

18 other genes

18 other nuclear genes

No qualifying regional + MAGMA signal

Total tau

GCST90726397

1

APOE

regional + gene-based signal



1 APOE

18 other genes

18 other nuclear genes

No qualifying regional + MAGMA signal

p-tau181

GCST90726398

1

APOE

regional + gene-based signal



1 APOE

18 other genes

18 other nuclear genes

No qualifying regional + MAGMA signal

B All 57 screens

3

APOE-positive
gate decisions

Same gene in all three traits



3

54

54

without qualifying
GWAS signal

0

newly supported
genes

Three positive decisions = APOE across three traits, not three genes

Regional + MAGMA gates do not establish a molecular mechanism or colocalization.